

Muhammad Ahmad

FA23-BCE-113

## Engineering Economics Assignment # 2

Q1:

Given

X: Economic life = 5 year

Annual cost = 4200

Y: Initial cost = Rs = 18,000

Salvage = S = 0

Study period = n = 5 year

interest =  $i = 12\% = 0.12$

Maintenance cost:

first year cost ( $A_1$ ) = 2000

Annual increment = 500

AEC total = CR + A<sub>maint</sub>

CR =  $P(A/P, i, n) - S(A/g, i, n)$

A<sub>maint</sub> =  $A + G(A/g, i, n)$

Using table:

$i = 12\%$  ,  $n = 5$

Capital Recovery factor =  $(A/P, 12\%, 5) = 0.2279$

Gradient Uniform Series factor =  $(A/g, 12\%, 5) = 0.775$



$$CR = P \times (A/P, 12\%, 5)$$

$$CR = 1800 \times 0.2774$$

$$CR = 499.32$$

Calculation of Equivalent annual cost

$$A_{\text{maint}} = 2000 + (500 \times 1.775)$$

$$A_{\text{maint}} = 2887.5$$

$$AEC_r = CR + A_{\text{maint}}$$

$$499.32 + 2887.5$$

$$AEC_r = 7880.70$$

now

machine X: annual eq. cost = 4200

machine Y: annual eq. cost = 7880

Cost to operate existing machine X is significantly lower than cost of purchase to operate the new machine Y. So replacing isn't a good option.